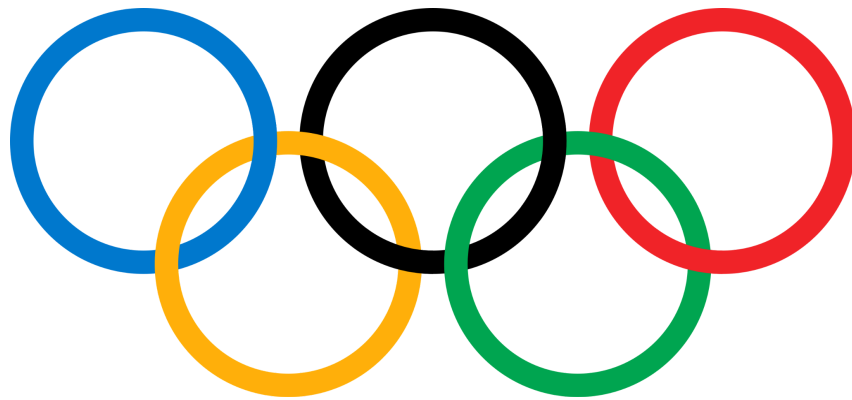


Olympic Data



Christina Grevy	4127286	chgre23@student.sdu.dk	Percent of Contribution: 100%
Maria Tuesen	201834942	matue23@student.sdu.dk	Percent of Contribution: 100%
Published Shinyapp:	https://christinagrevy.shinyapps.io/OlympicDataProject/		
Github Repository:	https://github.com/cgrevy/Olympic		
Dataset:	https://www.kaggle.com/datasets/bhanupratapbiswas/olympic-data/data		
AI-generated graph:	https://chatgpt.com/share/693170f6-ca84-800d-ad09-f986677cb1b6		



University of Southern Denmark

BSc Software Engineering

5. Semester 2025

Abstract	3
1. Background and Motivation.	4
2. Project Objectives	5
3. Data	6
4. Visualization/Dashboard:	7
5. Story/Results	8
6. Conclusion/Discussion	13

Abstract

The goal of the project is to visualize and get an understanding of how the Olympic Games have evolved over the years. The Olympic Games have changed a lot since they started in 1896 and have gained a lot of popularity. A lot of new sports has been added and the number of participants has increased throughout the years. Our main focus is to explore how the size of the Games, the gender distribution and the athletes physical attributes have changed since the Games started in 1896.

The visualizations are built on the dataset “Olympic Data”, which is a dataset covering the international multisport event held every four years. The dataset covers the name, gender, height, weight and age of the athlete, the year and sport they participated in, the team they represented, what committee they were represented by, whether they participated in Summer or Winter Olympics and what medal they won, if any.

1. Background and Motivation.

The Olympic Games creates a worldwide gathering point for the entire population. It is an event where people can lay their differences aside and come together in celebrating sports and achievements. But how have the Olympics evolved over time? And what can we learn from the data of its participants?

Coming from sports backgrounds we have a personal interest in human performance and competition. It is interesting to see which metrics play a role in winning medals and producing top athletes, and how these differ between sports.

Is an athlete just an athlete? Different sports favor different attributes, and how does this look? We are interested in comparing what attributes play a part in winning medals, and see how metrics such as age, weight etc. play a role in reaching the top step of the podium. We also want to compare these metrics to how it was when it started.

2. Project Objectives

To guide our visualization we have formulated the following project objectives:

1. Which countries have won the most medals over time, and is there a difference between the summer and the winter games?
2. How has the gender distribution changed from 1896 to 2016?
3. How old are the Olympic athletes?
 - a. Which sport has the highest average age?
 - b. Which sport has the lowest average age?
 - c. How old were the participants from 1896 to 2016?
4. Is there a correlation between the athletes physical appearance and their sport, and how does this contribute to the number of medals won?
 - a. Is there a correlation between the athletes height and the number of medals they win?
 - b. Is it different depending on the sport?
 - c. Is there a correlation between the athletes height and weight depending on the sport?
5. Which countries have hosted the games?
 - a. How many participants were in the games?
6. How many sports have been a part of the games?
 - a. How many participants have been part of each sport over the years?
7. How many different countries have been represented at the games?
8. Who are the most winning athletes?

3. Data

The dataset has been obtained from the website Kaggle¹ and contains 35.658 rows with records of athletes. Each row represents an athlete at each of the games. So if the same athlete competes in multiple games and/or events, they will appear multiple times in the data.

The dataset covers the name, gender, height, weight and age of the athlete, the sport and event they participated in, what medal they won (if any), the team (Country) they represented, what committee they were represented by, whether they participated in the Summer or Winter Olympics and the hosting city.

When working with the dataset we noticed that some athletes were missing. For example, Michael Phelps is not included in the dataset, even though he is the most winning Olympic athlete ever.

¹ <https://www.kaggle.com/datasets/bhanupratapbiswas/olympic-data>

4. Visualization/Dashboard:

Design

For our graphs we have chosen to split them up into the categories: Size of the Games, Top Winning Countries, The Physics of an Olympian, Age in the Olympics and Most Winning Athletes.

In each of these categories we have displayed the graphs that relate to our project objectives. To achieve a cohesive visual aesthetic all the graphs are in the Olympic colors, relating to the theme of the project.

Must Have Features

We have made a list of some of the features that is a must to include before we can say that the project is finished:

1. At least three types of graphs.

Some of the graphs we have chosen to include are a barchart, a histogram, a scatterplot with lines connecting the dots and a time series graph.

2. At least one animated graph.

We have decided to include an animated graph as a scatter plot with lines connecting the dots and an animated time series graph.

3. An AI-generated graph.

We want to try and get ChatGPT to make a graph for us. When doing this, we do not want to specify which type of graph it should make but just give it the dataset and what project objective to answer and see what it comes up with.

4. In total, at least nine graphs.

Before we start making the graphs, we have made a list of what graphs we will make for each of the project objectives, this list includes eleven graphs in total.

5. A link to the dashboard/visualisation.

6. An option to download the report.

5. Story/Results

The following section will answer the research questions we asked in section 2, using the graphs we described in section 3.

1. Which countries have won the most medals over time, and is there a difference between the Summer and the Winter Games?

This question has been answered with the visualisations found in the “Top Winning Countries” tab.

When answering this question we expected the United States to be the most winning country, however we were surprised to see how close France and Italy came second in respectively the Summer and Winter Games. When looking at the barchart displaying the top winning countries in the history of the Olympics, we can see that the top 3 medal winning countries in both the Summer and Winter Games are the United States, France and Italy.

When toggling with the time period to look at, we can see that if we only look at the last 30 years new countries appear in the top 15, like China, Romania and Russia. However, many of the countries in the top 15 in the overall history of the Games are the same as the last 30 years and the first 30 years. This could indicate the significance of a continuous participation in the Games, since the beginning.

We created a line graph to show the top 5 countries’ amount of medals won per game. It would however also be interesting to have created a graph showing the development of the amount of accumulated medals per country.

2. How has the gender distribution changed from 1896 to 2016?

This question has been answered with the two area charts at the bottom of the page in the “*Size of the Games*”-tab.

The two area charts each draw the percentage of men and women participating in the Summer and Winter Games. Both charts show a positive trend of women participation in the game, suggesting a development towards equal distribution of male and female participants.

In 2016 women represented 43,5% of the athletes in the Summer Games, compared to the Games in 1924 where women only represented 5,5%. In 2014 women accounted for 39,9% of the participants in the Winter games, compared to the 2,6% in 1924.

3. How old are the Olympic athletes?

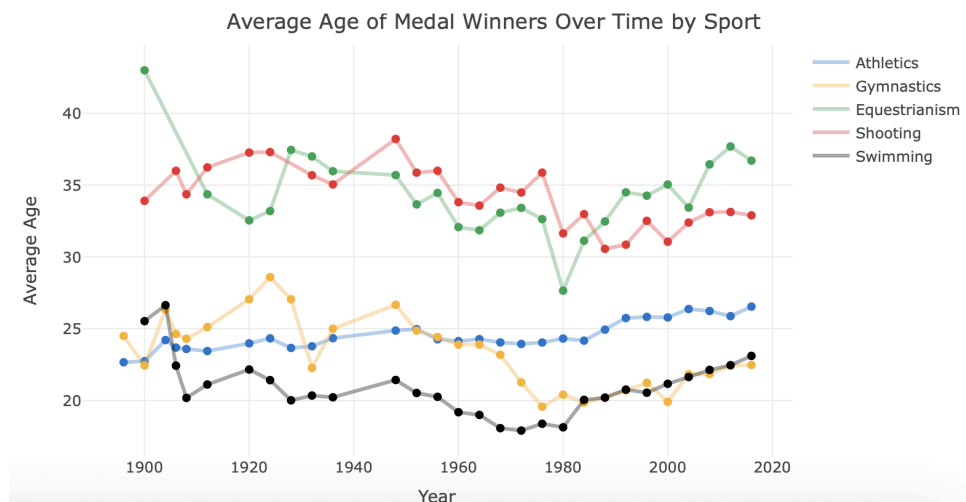
- a. Which sport has the highest average age?**
- b. Which sport has the lowest average age?**
- c. How old were the participants from 1896 to 2016?**

This question and its subquestions have been answered with the visualisations found in the “*Age in the Olympics*” tab.

The sport with the highest average age is Arts Competition with an average age of 45.7 years, the sport with the lowest average age is Rhythmic Gymnastics with an average age of 18.6 years. We can see from the bar chart that most of the sports have athletes with an average age that is younger than the overall average age which is 27 years.

On the graph “*Distribution of Age per Sport*”, the distribution of age per sport is shown with a histogram where the ages are on the x-axis and the number of athletes are on the y-axis. On the left side, you can choose between the sports and the year range. From this graph we see that most of the participants are in their 20s, with the highest number of participants at age 23. From this we can also see that the youngest participant ever was 11 years old and participated in Figure Skating.

To compare the medal winners’ age with each other, we have created a time series graph:



This shows a higher average age for medal winners in certain sports like Equestrianism and Shooting, where other sports like Gymnastics, Athletics and Swimming tend to have much younger medal winners.

4. Is there a correlation between the athletes physical appearance and their sport, and how does this contribute to the number of medals won?

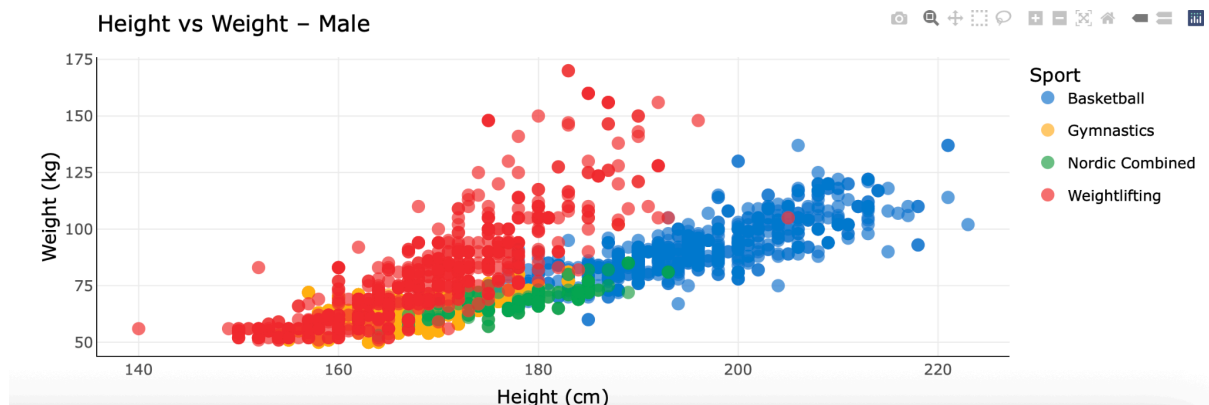
- a. Is there a correlation between the athletes height and the number of medals they win?
- b. Is it different depending on the sport?
- c. Is there a correlation between the athletes height and weight depending on the sport?

The visualisations relating to question 4 and its subquestions are in the “*The Physics of an Olympic*” tab.

The scatter plot “*Height of Athletes Correlation with Medals Won*” has height on the x-axis and medal count on the y-axis. One sport and gender can be seen at the time. The majority of the sports has a bell shaped distribution with a few outliers. It is hard to see a direct correlation between a certain height and the number of medals won. However there is a difference between the most common height amongst medal winners in the different sports. For example Basketball has much taller athletes than Gymnastics.

When looking at the physiques of the athletes, we chose to compare their height and weight ratio, between different sports. While some sports didn’t have a strong trend, others had very

clear patterns. When comparing Weightlifting athletes with Gymnastics athletes, we can see they have fairly similar heights, however weightlifters tend to be heavier than the gymnasts with the same height. Basketball players are among the tallest athletes. Nordic Combined has a bit taller athletes than gymnastics, however most of them are quite light.



5. Which countries have hosted the games?

a. How many participants were in the games?

To gain an overview of all the host cities of the Olympic Games, we created a timeline seen in the “*Size of the Games*” tab. When the first (modern) Olympic Games were held, there were 34 participants. This number has grown into 3003 participants in the 2016 Games in Rio de Janeiro.

6. How many sports have been a part of the games?

a. How many participants have been part of each sport over the years?

For each bubble on the timeline, it is possible to see more details on the Olympic Game that year. Here you can see the number of sports that have been a part of the game, as well as a list of unique sports in the games. Unique sports are defined by having only been part of one Olympic Games.

To see the number of participants who have partook in each sport, see the time series graph, also in the “*Size of the Games*” tab. The graph shows that some sports, like Swimming and Athletics have had an increasing number of participants for each game, while others like Gymnastics and Equestrianism have had a quite stable number of participants since the 1960s circa.

To further improve the timeline visualisation, the graph could get a “highlight” section that sum up the number of different sports that have been part of the Olympic Games over the years, the country that has hosted the games the most times, and the total number of different athletes that have been part of the Olympic Games, i.e. how many Olympians exist?

7. How many different countries have been represented at the games?

This question has been answered with a graph in the “*Size of the Olympic Games*” tab. The graph has the Games/year on the x-axis and the number of participating countries on the y-axis. The Summer Games are represented by a yellow line and the Winter Games are represented by a blue line. Looking at the graph we can see that the Games have gotten more participating countries over the years. The Summer Games started in 1896 with 7 countries participating. This number has grown to 180 countries participating in 2016. The Winter Games started in 1924 with 16 countries participating and has grown to 62 countries participating in 2014.

A possible explanation as to why the Summer Olympics have more than double the amount of participating countries compared to the Winter Olympics could be that the winter sports are not as accessible to train in every country, compared to the summer disciplines.

9. Who are the most winning athletes?

To find the most winning athlete, we let an AI generate a visualisation. This visualisation can be seen in the tab “*Most Winning Athlete (AI)*”.

To create the visualisation we gave ChatGPT the prompt:

“We have this olympic dataset:

<https://www.kaggle.com/datasets/bhanupratapbiswas/olympic-data>

can you in R make something that visualizes the most winning athletes?”

ChatGPT then created an R script that makes a graph where you can select the number of top athletes and medal type to see. It shows the name and amount of medals won, also including a table where you can see the name of the athlete, the total medals won and the amount of gold, silver and bronze medals.

While seeing the results we realised here that the dataset is missing some athletes like the top winning athlete Michael Phelps.

In general we were quite pleased with the visualisation that ChatGPT created. It provided a nice overview, with nice details like the different medals being shown in different colors on the stacked barchart. To further improve the visualisation we would like to add more human friendly hover texts and labels, as well as provide more details, like which sport and how many Games the top winning athletes have participated in.

6. Conclusion/Discussion

For this project, we wanted to explore how the Olympic Games have evolved from their early years to today, focusing on athlete physics, gender distribution, participation levels, and medal-winning patterns. By using the Olympic dataset from Kaggle and creating a range of visualizations, we were able to see trends across the history of the Olympics.

Our visualizations showed a significant expansion of the Games over time, with more participating countries, more athletes, and a broader range of sports. Gender representation has shifted dramatically, progressing from underrepresentation of women to almost equal amount of men and women in 2016, reflecting social changes. We also found that age patterns vary widely across sports, with most athletes clustered in their 20s, while specific disciplines, like Art Competitions, drew much older participants.

By examining physical characteristics, we found predictable correlations between height, weight, and sport-specific demands. Our graphs confirmed that different sports favor different body types, and while height appears to influence medal counts in some disciplines, this relationship is highly sport-dependent.

One of the challenges in the project was that the dataset was not fully complete. We noticed late in the project that some of the bigger athletes like Micheal Phelps were missing. This is an indication that the dataset is incomplete or curated inconsistently, which also affects the accuracy of the graphs.

Another challenge in the project was that some athletes were missing values like weight and height, especially in the earlier years. This results in some of the graphs being biased towards the more recent Games where the data collection is more complete.

In conclusion, our work provides a clear overview of how the Olympics have changed over time. By combining analytical methods, and interactive visualizations, we gained answers to our research questions.